# Week 12: Maps

November 11, 2026

## WHY MAPS ARE RISKY

A map is a chart that uses geography as its coordinate system. That brings two problems no other chart has. The earth is round and your screen is flat, so every map **distorts** something. And geographic area has nothing to do with the thing you're mapping, so a huge empty state gets lots of ink and a dense city gets almost none.

Maps are also the easiest chart to make look authoritative while saying nothing. A map that renders is not a map that is right.

### First, ask: do you need a map?

If the location isn't the point, a bar chart of the same data is often clearer. Make both, and say so when the bar chart wins.

> **Land doesn't vote, people do.** A map filled by a raw count mostly shows where the people live. Divide by population before you trust the pattern, and check how many places actually got filled.

## STEP 1: GET A SHAPE FILE

A shape file is a data frame where each row is a place and one column holds that place's geometry. Two flavors, two geoms.

| Format | Get it from | Plot with |
|---|---|---|
| Polygon data | `map_data()` | `geom_polygon()` |
| Simple Features (sf) | `rnaturalearth`, `tigris`, `maps`, `st_read()` | `geom_sf()` |

### Polygon maps

```
world <- map_data("world")

ggplot(world) +
  geom_polygon(
    aes(x = long, y = lat, group = group),
    fill = "grey90", color = "grey60"
  )
```

Forget `group = group` and the polygons connect into spaghetti.

### Simple Features maps

```
library(rnaturalearth)

world <- ne_countries(
  scale = "medium", returnclass = "sf")

ggplot(world) +
  geom_sf(fill = "grey90", color = "grey60")
```

| Need | Use |
|---|---|
| Countries | `ne_countries()` |
| States or provinces | `ne_states(country = ...)` |
| US with AK and HI moved | `tigris::states()` then `shift_geometry()` |
| US counties | `st_as_sf(map("county", plot = FALSE, fill = TRUE))` |
| A shape file on disk | `st_read(file.path('data', 'x.shp'))` |

## STEP 2: ADD DATA

### Choropleth: fill regions by a variable

Join your data onto the shape file, then map the variable to `fill`.

```
us_states <- ne_states(
  country = 'united states of america',
  returnclass = 'sf') %>%
  left_join(milk_2017, by = 'name')

ggplot(us_states) +
  geom_sf(aes(fill = milk_produced)) +
  scale_fill_viridis()
```

**The join is where maps break.** Names that don't match drop out silently and leave grey holes on a map that still renders happily.

| Culprit | Example |
|---|---|
| Different spellings | "Washington, D.C." vs "District of Columbia" |
| Different codes | `USA` vs `United States` |
| Stripped leading zeros | FIPS code `01001` becomes `1001` |

Check by counting: rows in, filled places out. Don't eyeball it.

Fill scale breaks can change the story, so try several. If big empty areas dominate, consider a hex or tile map.

### Points: locations and sizes

Add a point layer on top of the shape file. Points can mark locations or encode a variable.

```
ggplot() +
  geom_sf(data = us_states_cont) +
  geom_point(
    data = us_coffee_shops,
    aes(x = long, y = lat, color = name),
    size = 0.3
  ) +
  theme_void()
```

Point data in lat/long still needs `x = long, y = lat`. Filter to the region first, or stray points stretch the map.

When size encodes a value, map it to **area**, not radius. Area is $\pi r^2$, so doubling the radius makes the point four times as big.

## PROJECTIONS

A projection is how the round earth gets flattened. Every one distorts area, shape, distance, or direction, and none is "best". It depends on the claim your map is making.

Set it with one layer:

```
ggplot(world) +
  geom_sf() +
  coord_sf(crs = "ESRI:54030")
```

| World | Code |
|---|---|
| Robinson (good default) | `"ESRI:54030"` |
| Mercator (inflates the poles) | `"ESRI:54004"` |
| Mollweide (equal area) | `"ESRI:54009"` |

| United States | Code |
|---|---|
| Albers (good default) | `"ESRI:102003"` |
| Lambert Conformal Conic | `"ESRI:102004"` |

## CHECKING AN AGENT'S MAP

| Check | Ask |
|---|---|
| Join | How many places were filled, and which didn't match? |
| Measure | Is this a count or a rate? |
| Projection | Does it distort the areas the claim is about? |
| Format | Would a bar chart have been clearer? |

## BEFORE NEXT CLASS

- **HW11: Interactivity**, due **Nov 16**.
- **Mini Project 3: Re-design**, due **Nov 22** (9% of grade).

You should be able to:

- Load a shape file as polygon data or as sf, and pick the matching geom.
- Join data to a shape file and count the places that failed to match.
- Make a choropleth and a point map with a sensible color or size scale.
- Explain why a raw-count choropleth is often just a population map.
- Change a map's projection with `coord_sf()` and say what it distorts.